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HTExploit – Bypassing .htaccess restrictions

[image: HTExploit !]

HTExploit is a tool that exploits a vulnerability in the way that the .htaccess files are configured in an Apache Web server (and also in many of its derivatives)

In September, 1997, an interesting article appeared in Apacheweek. This article commented about a design flaw in the way that the < *Limit* > tags were handled, that allowed to jump certain restrictions.

By having this configuration:

```
< Limit >
```

```
... restricciones de acceso como 'require' o 'allow'...
```

```
< /Limit >
```

All the traffic that uses the GET and POST methods had to go through the access controls. And if any method other than GET and POST were used, the communication would be automatically dropped, because an unauthorized method would have been invoked.

But: What happens if we use a non-standard method?

In this case, our Apache server will automatically decide to send the request to the software running on the upper layer, so that it can handle the request.

Here is where it gets tricky:

If the agent above Apache is PHP (in any version, up to what I found in my investigation), and if it receives a non-standard method, PHP will treat it like a GET and process it.

What does all this mean?

We can invoke a GET on the remote server WITHOUT USING GET, bypassing this way the restrictions applied in the .htaccess file.

To sum it up, this is what the tool exploits:

[[image: image]](http://www.matiaskatz.com/wp-content/uploads/2013/01/htexploit.png)

The reach of the tool is pretty huge. Basically, with any Apache+PHP combination on the server, and with an .htaccess file configured in the way previously shown, **we can exploit the server.**

There are at least 3 simple ways to mitigate this problem:

- Using the < *LimitExcept* > *tags: By applying this tag inside the .htaccess file, we can indicate Apache how to react when presented with a method not indicated in the *< *Limit* > tag
- Applying controls in the code: By verifying that the variable *$PHP_AUTH_USER* is set, we can assure that the user has been authenticated by the Webserver. Also, we can check that the variable *$_SERVER["REQUEST_METHOD"]* is either GET o POST, and drop any other method
- Restricting the methods from Apache: By using the famous *modsecurity* module, we can apply the *allowed_methods* policy, and that way filter the methods that will be allowed by the Web server

The tool was presented at Black Hat USA 2012 and is publicly available since July 25th, 2012.

Its latest version is 0.77. You can find it on the project's web site, or in Backtrack 5 R3

It's Open source, and it has been developed in Python, so you will be able to use it under any OS. It doesn't require any special Python module.

On the project's website you will also find more information about the root of the vulnerability, and will be able to see in detail how to use the tool.

If you want to read more about the project and the tool, you can read the official Black Hat paper.

Here are some references that can help you investigating the issue:

- Apache Tutorial: .htaccess files:

http://httpd.apache.org/docs/2.0/howto/htaccess.html

- Common Configuration Problems: Issue #81

http://www.apacheweek.com/issues/97-09-05#configerrors

- HTTP Authentication: Basic and Digest Access Authentication

http://tools.ietf.org/html/rfc2617

- Authentication, Authorization and Access Control

http://httpd.apache.org/docs/2.4/howto/auth.html

- HTExploit project Website:

http://www.htexploit.org
